



KAMPALA INTERNATIONAL UNIVERSITY – WESTERN CAMPUS
DEPARTMENT OF PHARMACOLOGY AND TOXICOLOGY
BASIC PHARMACOLOGY EXAM FOR BMS 1.2

DATE: 23rd Jan 2021

DURATION: 3 hours

Instructions

Answer ALL questions in section A using the special MCQ template provided and only two questions from section B using the booklet provided.

1. The most important factor which governs diffusion of drugs across capillaries other than those in the brain is:
 - A. Blood flow through the capillary
 - B. Lipid solubility of the drug
 - C. pKa value of the drug
 - D. pH of the medium
2. Active transport of a substance across biological membranes has the following characteristics except:
 - A. It is specific
 - B. It is pH dependent
 - C. It is saturable
 - D. It requires metabolic energy
3. Drugs which undergo high degree of first-pass metabolism in liver:
 - A. Have low oral bioavailability
 - B. Are excreted primarily in bile
 - C. Are contraindicated in liver disease
 - D. Exhibit zero order kinetics of elimination
4. Glomerular filtration of a drug is affected by its:
 - A. Lipid solubility
 - B. Plasma protein binding
 - C. Degree of ionization
 - D. Rate of tubular secretion
5. A patient is prescribed a drug that has an extremely short half-life for chest pain (angina). Which modification in drug dosage or scheduling should the nurse do?
 - A. Once-daily dosing
 - B. More frequent dosing
 - C. Larger first dose and smaller repeat doses
 - D. Parenteral doses greater than enteral doses
6. Which of these factors will affect the patient's bioavailability after taking an oral dose of ceftriaxone when treating their peptic ulcers?
 - A. Increased time of gastric emptying
 - B. Increased lipid solubility of drug
 - C. Increased hepatic metabolism of a drug
 - D. Increased rate of drug dissolution
7. A patient is experiencing toxic drug reaction. An intern doctor remembers that the margin of safety of the drug is low. All of the following would decrease its margin of safety except:
 - A. Renal failure
 - B. Hepatic impairment
 - C. Increase in clearance
 - D. Increase in neurotoxicity
8. After a patient getting an intravenous bolus injection of lidocaine, the initial plasma concentration changes. What is the most outstanding factor affecting this initial plasma concentration?
 - A. Dose & apparent volume of distribution
 - B. Dose & clearance
 - C. Apparent volume of distribution & clearance
 - D. Clearance & half-time
9. The pharmacokinetic value that most reliably reflects the amount of drug in tissues whose value can be used to interpret the amount of drug in tissues:
 - A. Peak blood concentration

- B. Time to peak blood concentration
 - C. Product of the V_d and the first-order rate constant
 - ~~D. Volume of distribution V_d~~
10. A patient is administered to the ER for treatment of drug overdose. The identity of the drug is unknown, but it is observed that when the urine pH is acidic, the renal clearance of the drug is less than the glomerular filtration rate, and when the urine pH is alkaline, the clearance is greater than the glomerular filtration rate. The drug probably is:
- A. Strong acid
 - B. Strong base
 - C. Weak base
 - D. Weak acid
11. A competitive antagonist is a substance that:
- A. Interacts with receptors and produces submaximal effect
 - ~~B. Binds to the same receptor site and progressively inhibits the agonist response~~
 - C. Binds to the nonspecific sites of tissue
 - D. Binds to one receptor subtype as an agonist and to another as an antagonist
12. A partial agonist can antagonize the effects of a full agonist because it has:
- ~~A. High affinity but low intrinsic activity~~
 - B. Low affinity but high intrinsic activity
 - C. No affinity and low intrinsic activity
 - D. High affinity but no intrinsic activity
13. Receptor agonists possess:
- ~~A. Affinity but no intrinsic activity~~
 - B. Intrinsic activity but no affinity
 - C. Affinity and intrinsic activity with a + sign
 - D. Affinity and intrinsic activity with a - sign
14. The receptor transduction mechanism with the fastest time-course of response effectuation is:
- A. Adenylyl cyclase-cyclic AMP pathway
 - ~~B. Phospholipase C-IP₃: DAG pathway~~
 - C. Intrinsic ion channel operation
 - D. Protein synthesis modulation
15. Mark out the substances whose mechanisms are based on interaction with ion channels
- A. Sodium channel blockers
 - B. Calcium channel blockers
 - C. Potassium channels activators
 - ~~D. All of the above~~
16. The following statement is not true of log dose-response curve:
- A. It is almost linear except at the ends
 - B. It is a rectangular hyperbola
 - C. It facilitates comparison of different agonists
 - ~~D. It can help in discriminating between competitive and noncompetitive antagonists~~
17. What does "affinity" mean?
- A. A measure of how tightly a drug binds to plasma proteins
 - ~~B. A measure of how tightly a drug binds to a receptor~~
 - C. A measure of inhibiting potency of a drug
 - D. A measure of bioavailability of a drug
18. If an agonist can produce submaximal effects and has moderate efficacy it's called:
- ~~A. Partial agonist~~
 - B. Antagonist
 - C. Agonist-antagonist
 - D. Full agonist
19. Tick the second messenger of G-protein-coupled (metabotropic) receptor:
- ~~A. Adenylyl cyclase~~
 - B. Sodium ions
 - C. Phospholipase C
 - D. Camp
20. The antidotal action of sodium nitrite in cyanide poisoning is based on:
- A. Physical antagonism
 - B. Chemical antagonism
 - C. Physiological antagonism
 - D. Noncompetitive antagonism

21. The pharmacokinetic parameter which determines the speed of drug input that must balance the speed of drug elimination to achieve a steady state concentration
- A. Clearance
 - B. Dosing rate
 - C. Bioavailability
 - D. Volume of distribution
22. What would be a patient's response to a drug dose that is below the minimum effective concentration (MEC)?
- A. Drug entry exceeds drug elimination.
 - B. The risk for toxic side effects is increased.
 - C. The drug's duration of action is longer than expected
 - ☒ D. The intended response fails to occur.
23. A patient receives a drug through the intravenous (I.V) route of administration. Duration of action of the I.V administered drug depends on
- A. Protein binding
 - B. Clearance
 - C. Distribution volume
 - D. Lipid solubility
24. Kamuzu took a tablet of a painkiller after feeling a headache. Which of these equations below best describes the rate of drug dissolution of a tablet?
- A. Michaelis – Menten equation
 - ☒ B. Henderson – Hassel Bach equation
 - C. Fick's law
 - D. Law of mass action
25. Which of these characteristics explains phenytoin as a drug with nonlinear pharmacokinetics?
- A. A constant ratio of drug metabolites is formed as the administered dose increases
 - ☒ B. The elimination half-life ($t_{1/2}$) increases as the administered dose increases
 - C. The area under the plasma drug concentration versus time curve (AUC) increases in direct proportion to an increase in the administered dose
 - ☒ D. Both low and high doses follow first-order elimination kinetics
26. All the following statements about unwanted effects of drugs (adverse reactions) are correct EXCEPT:
- ☒ A. There is a fundamental biological distinction between therapeutic and adverse effects of drugs.
 - B. Some adverse effects are due to normal predictable pharmacological effects of drugs and may occur in any patient taking the drug.
 - C. Some adverse effects are due to abnormalities in the patient and will only occur in some patients taking the drug.
 - D. The terms intolerant and tolerant refer to individuals at either extreme of the normal distribution curve.
27. For a drug eliminated by a first-order kinetic process:
- A. A constant amount of drug is eliminated per unit time
 - B. The duration of action will be longer than for a drug eliminated by zero-order process.
 - C. The elimination rate constant will be independent of the dose administered.
 - D. It is unlikely to bind significantly to plasma albumin.
28. A patient received on emergency ward with a severe infection was given a loading dose. What does the Loading dose of a drug primarily depend on?
- A. Volume of distribution
 - B. Clearance
 - ☒ C. Rate of administration
 - D. Half life
29. All of the following would decrease the therapeutic index of a drug except:
- A. Renal failure
 - B. Hepatic impairment
 - ☒ C. Increase in clearance
 - D. Increase in neurotoxicity
30. Concerning the effect of pH on the urinary excretion of drugs, it can be correctly stated that
- A. Urinary acidification accelerates excretion of weak acids and bases
 - B. Urinary alkalization accelerates excretion of weak acids and bases
 - C. Urinary acidification accelerates excretion of weak acids

- D. Urinary alkalization accelerates excretion of weak acids
31. Tick the substances whose mechanisms are based on interaction with ion channels
- A. Sodium channel blockers
 - B. Calcium channel blockers
 - C. Potassium channels activators
 - ☒ D. All of the above
32. If two drugs with the same effect, taken together, produce an effect that is equal in magnitude to the sum of the effects of the drugs given individually, it is called as:
- A. Antagonism
 - B. Potentiation
 - ☒ C. Additive effect
 - D. None of the above
33. What does the term "potentiation" mean?
- A. Cumulative ability of a drug
 - B. Hypersensitivity to a drug
 - ☒ C. Fast tolerance developing
 - D. Intensive increase of drug effects due to their combination
34. What phenomenon can occur in case of using a combination of drugs?
- A. Tolerance
 - B. Tachyphylaxis
 - C. Accumulation
 - ☒ D. Synergism
35. A teratogenic action is:
- A. Toxic action on the liver
 - B. Negative action on the fetus causing fetal malformation
 - ☒ C. Toxic action on blood system
 - D. Toxic action on kidneys
36. Idiosyncratic reaction of a drug is:
- A. A type of hypersensitivity reaction
 - B. A type of drug antagonist
 - C. Unpredictable, inherent, qualitatively abnormal reaction to a drug
 - D. Quantitatively exaggerated response
37. Factors that may cause variation in drug responsiveness:
- ☒ A. Changes in the number or function of receptors
 - B. Idiosyncratic drug responses
 - C. Hypersensitivity reactions
 - D. All of the above
38. The term "chemical antagonism" means that:
- ☒ A. Two drugs combine with one another to form an inactive compound
 - B. Two drugs combine with one another to form a more active compound
 - C. Two drugs combine with one another to form a more water-soluble compound
 - D. Two drugs combine with one another to form a more fat-soluble compound
39. An example of an agent that exerts much of its effects through intracellular receptors that in complex form binds to DNA response elements:
- A. Acetylcholine
 - B. Dopamine
 - C. Corticosteroids
 - D. Diltiazem
40. 'Drug efficacy' refers to:
- A. The range of diseases in which the drug is beneficial
 - B. The maximal intensity of response that can be produced by the drug
 - C. The dose of the drug needed to produce half maximal effect
 - ☒ D. The dose of the drug needed to produce therapeutic effect
41. The rectal route of administration
- A. Provides nearly 100 percent bioavailability
 - B. Is only used for localized effects
 - ☒ C. Is suitable for persons with nausea and vomiting
 - D. Is subject to a high degree of first-pass metabolism
42. If a drug exhibits saturation (zero-order) kinetics, then
- A. The rate of drug elimination is constant

- B. Drug half-life is constant
C. Drug clearance is constant
D. Plasma drug concentration is constant
43. Systemic clearance may be expressed as:
A. L/hour or ml/per second
B. ml/hour
C. g/litre
D. Weight/volume
44. A patient is prescribed a drug that has an extremely short half-life for chest pain (angina). Which modification in drug dosage or scheduling does the nurse expect?
A. Once-daily dosing
B. More frequent dosing
C. Larger first dose and smaller repeat doses
D. Parenteral doses greater than enteral doses
45. Drug E is an acidic drug that will bind to plasma protein when in plasma. Which of these plasma protein binds Aspirin with the greatest?
A. Albumin
B. alpha 1- acid glycoprotein
C. Both (a) and (b)
D. Only (a)
46. ~~None of the above~~ Usually, during the safety study of a proposed drug purpose is to establish dosage levels below which toxicity is not observed and to identify effects on the vital organs in a healthy man:
A. Preclinical studies
B. Phase IA clinical studies
C. Phase IB Clinical studies
D. Post Marketing drug studies
47. Sub Chronic toxicity in rats is done
A. Usually done sub way between 14 and 21 days
B. Usually done sub way between 14 and 28 weeks
C. Usually done fortnightly beginning at day 14 and ending at day 40
D. None of the above
48. The following is an important question asked during one of the Clinical trial phases- (stages during drug development)?
A. Does the extract from this source have the alleged Pharmacological activity?
B. What are the chemicals responsible for the observed pharmacological activity?
C. Is it safe (As demonstrated by Pre-clinical and Clinical trials)?
D. How much will the finished product cost?
49. Drug Discovery and development usually begins with which stage?
A. Pre-clinical Phase
B. Research concept and proposal
C. Chemistry and Pharmacy formulation
D. None of the above mentioned
50. Regulatory clearance is in which stage of the drug development process?
A. Concept
B. Pre-clinical stage
C. Clinical trial
D. Launching
51. The following statement is true in relation to 'drug toxicity' and 'poisoning':
A. The two terms are synonymous
B. When a toxic effect requires specific treatment, it is called poisoning
C. A toxic effect which endangers life by markedly affecting vital functions is called poisoning
D. Toxicity is caused by drugs while poisoning is caused by other harmful chemicals
52. An immunologically mediated reaction to a drug producing stereotyped symptoms unrelated to its pharmacodynamic actions is:
A. Hypersensitivity
B. Supersensitivity
C. Intolerance
D. Idiosyncrasy
53. Drugs producing allergic reactions generally act as:
A. Complete antigens
B. Haptenes

- C. Antibodies
D. Mediators
54. The following allergic drug reaction is caused by circulating antibodies:
☒ A. Anaphylactic shock
 B. Angioedema
 C. Serum sickness
 D. Systemic lupus erythematosus
55. The type II, type III and type IV hypersensitivity reactions can be suppressed by:
 A. Adrenaline
 B. Antihistaminics
☒ C. Corticosteroids
 D. Sodium Cromoglycate
56. Which of the following is not an example of an inherited drug response trait that can be used as a pharmacogenomic tool?
☒ A. Prolonged neuromuscular blockade by normal doses of succinylcholine
 B. Neurotoxicity caused by normal dose of isoniazid
 C. Hemolytic anemia due to glucose-6-phosphate dehydrogenase deficiency.
 D. Deficiency of a conjugating system in neonates making them develop the gray baby syndrome from chloramphenicol.
57. All the statements below, about single nucleotide polymorphisms are correct except:
 A. Not all single nucleotide variations change the coded protein because the genetic code is redundant
 B. A single amino acid may be coded by more than one triplet therefore a change in one nucleotide does not always change the amino acid coded by the triplet.
 C. All single nucleotide variations change the coded protein because the genetic code is redundant
 D. Genetic polymorphisms sometimes produce heterozygotes with some selective advantages within a population.
58. The metabolism of all the following drugs are affected by well-known genetic polymorphisms except that of:
 A. Succinylcholine
 B. Isoniazid
 C. Butyrylcholinesterase
 D. Warfarin
59. Some patients when treated with Succinylcholine suffer from prolonged respiratory paralysis also referred to as succinylcholine apnea. The reason for this is:
 A. The surgeon sometimes forgets to treat them with the right dose of pseudocholinesterase enzyme which is responsible for metabolism of succinylcholine.
 B. The surgeon has treated them with an overdose of pseudocholinesterase enzyme which is responsible for metabolism of succinylcholine.
☒ C. These patients have genetically determined deficiency in pseudocholinesterase and succinylcholine is metabolized only half as rapidly as in persons with normally functioning enzyme.
 D. The persons affected have other autosomal recessive traits which account for pseudocholinesterase deficiency.
60. Phase II metabolism of isoniazid, occurs by N-acetylation of the drug to produce a more polar metabolite. A genetic defect in slow acetylators is caused by the synthesis of less of N-acetyl transferase enzyme. Which of the following is false about acetylation of isoniazid?
 A. The slow acetylator phenotype is inherited as an autosomal recessive trait and is race dependent.
☒ B. The slow acetylator phenotype is associated with higher incidence of isoniazid-induced peripheral neuritis and drug-induced autoimmune disorders compared to normal individuals or fast acetylators.
 C. This genetic defect mainly affects isoniazid metabolism but has little effect on its half-life.
 D. In rapid acetylators isoniazid has a short half-life.
61. Cholinomimetic drugs can be used for the treatment of all the following conditions EXCEPT:
 A. Closed angle Glaucoma
 B. Bradycardia
☒ C. Cobra bite
 D. Myasthenia gravis

62. Which of the following cranial nerves does not carry parasympathetic outflow?
 A. Oculomotor
~~B. Trochlear~~
 C. Facial
 D. Glossopharyngeal
63. Diagnosis of myasthenia gravis is done by using:
 A. Edrophonium
~~B. Neostigmine~~
 C. Succinylcholine
 D. Atropine
64. All of the following effects are seen with cholinergic muscarinic receptor stimulation except:
 A. Sweating
~~B. Rise in blood pressure~~
 C. Bradycardia
 D. Urination
65. Which of the following provides the best explanation for neostigmine being preferred over physostigmine for treating myasthenia gravis?
 A. It is better absorbed orally
 B. It has longer duration of action
 C. It has additional direct agonistic action on nicotinic receptors at the muscle end plate
 D. It penetrates blood brain barrier
66. A direct acting cholinomimetic that is lipid soluble and has been used in the treatment of glaucoma is:
 A. Acetylcholine
 B. Physostigmine
~~C. Pilocarpine~~
 D. Neostigmine
67. Drug of choice for treatment of acute organophosphate poisoning is:
~~A. Atropine~~
 B. Pralidoxime
 C. Neostigmine
 D. d-Tubocurarine
68. Which of the following is used in the treatment of atropine toxicity?
 A. 2-Pralidoxime
 B. Naloxone
 C. Flumazenil
~~D. Physostigmine~~
69. Accepted therapeutic indications for the use of antimuscarinic drugs include all of the following except:
 A. Hypertension
~~B. Motion sickness~~
 C. Parkinson's disease
 D. To produce mydriasis and cycloplegia
70. Inhibition of acetylcholinesterase activity is an important factor in the treatment of?
 A. Parkinson's disease
 B. Schizophrenia
 C. Alzheimer's disease
 D. Poisoning
71. Two days after an abdominal surgery, your patient's bowel sound is not audible. Using your knowledge of autonomic nervous system pharmacology, which of the following would you consider administering to your patient?
 A. Physostigmine
~~B. Bethanechol~~
 C. Pilocarpine
 D. Atropine
72. A patient is referred to you with Sjögren's syndrome. Which of the following would you prescribe for him?
 A. Atropine
 B. Pilocarpine
 C. Edrophonium
~~D. Pyridostigmine~~
73. As a physician in the military, your unit was deployed in South Sudan in a peace keeping mission. Your unit came under a surprise attack and all of a sudden, a majority of the service men started convulsing. Which of the following would you administer to your members immediately if you are to be economical?
~~A. Atropine~~
 B. Physostigmine
 C. Pralidoxime
 D. Adrenaline
74. Nine hours later, you discovered that your unit commander is missing and a search for him was launched. He was found 11 hours later alive although unconscious. Which of these pharmacological agents would you give him?
~~A. Atropine~~
 B. Physostigmine

- C. Pralidoxime
D. Adrenaline
75. You are in emergency department and a patient requiring ventilatory support through intubation is brought. Which of the following would be your drug of choice?
A. Atracurium
B. Suxamethonium
C. Donepezil
D. Hexamethonium
76. While in the emergency department, a patient requiring treatment with atropine was inadvertently overdosed. What drug can be used to reverse this effect.
A. Pirenzepine
B. Physostigmine
C. Edrophonium
D. Pilocarpine
77. Which of the following is a non-depolarising neuromuscular blocker?
A. Atracurium
B. Suxamethonium
C. Dextroamphetamine
D. Carbamazepine
78. Which of the following produce its effect by stimulating nicotinic receptors?
A. Cisatracurium
B. Mivacurium
C. Suxamethonium
D. Hemicholinium
79. After a patient getting an intravenous bolus injection of lidocaine, the initial plasma concentration changes. What is the most outstanding factor affecting this initial plasma concentration?
A. Dose & apparent volume of distribution
B. Dose & clearance
C. Apparent volume of distribution & clearance
D. Clearance & half-time
80. The pharmacokinetic value that most reliably reflects the amount of drug in tissues whose value can be used to interpret the amount of drug in tissues:
A. Peak blood concentration
B. Time to peak blood concentration
C. Product of the V_d and the first-order rate constant
D. Volume of distribution V_d
81. The principle mechanism of action of one of these agents involves the inhibition of reuptake of the catecholamines from the synaptic cleft of the adrenergic nervous system.
A. Monoamine oxidase inhibitors
B. Selective serotonin inhibitors
C. Tricyclic antidepressants
D. All the above
82. The principle mechanism of action of one of these agents involves the inhibition of reuptake of serotonin from the synaptic cleft of the serotonergic neurons.
A. Monoamine oxidase inhibitors
B. Selective serotonin inhibitors
C. Tricyclic antidepressants
D. All the above
83. The principle reason why adrenaline is not considered a neurotransmitter is
A. Producing a non-selective effect on the adrenergic system
B. Not secreted into blood as other neurotransmitters
C. Not secreted by an adrenergic neuron to effector organ
D. Is not a catecholamine
84. Tamsulosin is selectively used in the treatment of BPH among the adrenergic antagonist because?
A. It's the most potent α -1 blocker.
B. Selective binds α -1 A and B receptors of the prostate
C. Selectively binds to α -1 D receptors of the prostate
D. None of the above
85. Which of the following post-synaptic adrenergic neurons release acetylcholine as their neurotransmitters?
A. Those to the sweat glands
B. Those to the adrenal glands
C. Those to the walls of blood vessels
D. A and B are correct.

86. A 24-hour urine test was done for Mr. Kulambako and was found to have a higher urine metabolite of catecholamines. As the assistant to the consultant in the medical ward where you are attached, the consultant instructed you to put your patient on the best possible treatment before any other decision can be arrived at. Which of the following would you consider prescribing for your patient?
- A. Pheochromocytoma
B. Phentolamine
C. Phenoxybenzamine
D. Phenoxymethyl penicillin
87. First-dose phenomenon is a common side effects associated with the use of?
- A. Terazosin
B. Tamsulosin
C. Prazosin
D. Doxazosin
88. Which of the following would be a drug of choice for your patient with acute asthmatic attack?
- A. Inhaled steroid
B. Inhaled beta-2 agonist
C. Intravenous adrenaline
D. All the above
89. Which of the following is a long acting beta-2 agonist?
- A. Salbutamol
B. Albuterol
C. Terbutaline
D. Salmeterol
90. A patient report to a clinic where you are attached. She complains of serious attack at night. Which of the following would be your drug of choice for this patient?
- A. Salbutamol
B. Albuterol
C. Terbutaline
D. Salmeterol
91. A patient is brought to the emergency department where you are currently attached and he has both hypertension and BPH. Basing on your knowledge of ANS, which of the following drugs would you consider appropriate for this patient.
- A. Prazosin
B. Terazosin
C. Doxazosin
D. Tamsulosin
92. Which of the following beta antagonist is also known to block potassium channel and is used as an antiarrhythmic agent?
- A. Propranolol
B. Timolol
C. Sotalol
D. Atenolol
93. Which of the following beta antagonist can be easily used in certain form of heart failure?
- A. Propranolol
B. Nadolol
C. Carvedilol
D. Metoprolol
94. An adrenergic antagonist of choice for the treatment of somatic manifestation of anxiety is
- A. Prazosin
B. Propranolol
C. Phentolamine
D. All the above
95. Increased production of vanillylmandelic acid above daily normal limit is indicative of?
- A. Parkinsons disease
B. Alzheimer's disease
C. Pheochromocytoma
D. Cushing's syndrome
96. Which of the following is the rate limiting step in the synthesis of noradrenaline?
- A. Conversion of dopa to dopamine
B. Conversion of dopamine to noradrenaline
C. Conversion of tyrosine to dopa
D. Conversion of tyrosine to dopamine
97. Which of the following adrenergic neurotransmitter is important in the treatment of Parkinson's disease?
- A. Dobutamine
B. Dopa
C. Dopamine
D. Domperidone
98. Which of the following adrenergic receptors is important in the treatment of Parkinson's disease?
- A. α_1 receptor
B. β_1 receptors
C. D_2 receptors
D. None of the above
99. Which of the following adrenergic receptors is important in the treatment of schizophrenia?

- A. α_1 receptor
 B. β_1 receptors
 100. Which of the following is the none selective beta antagonist
 A. Atropine
 B. Propranolol
 C. D₂ receptors
 D. None of the above
 C. Ipratropium
 D. All the above

SECTION B

Choose only two questions from this part

1. In a tabular format, (25marks)

- Classify adrenergic drugs giving one example in each case. (6 marks)
- Describe briefly the mechanism of action of each group (8 marks)
- List 2 common clinical uses of each group with a drug example (6 marks)
- List 2 common side effects of each group excluding nausea, vomiting and diarrhea. (3 marks)
- List 2 contraindications of each group with a drug example. (2 marks)

2. In a tabular format, (25marks)

- Classify cholinergic drugs giving one example in each case. (6 marks)
- Describe briefly the mechanism of action of each group (8 marks)
- List 2 common clinical uses of each group with a drug example (6 marks)
- List 2 common side effects of each group excluding nausea, vomiting and diarrhea. (3 marks)
- List 2 contraindications of each group with a drug example. (2 marks)

3. Write short notes on the following pharmacological concepts, using examples where appropriate:

- Free and bound drug theory (5 Marks)
- Glomerular filtration of Drugs (5 Marks)
- Zero Order Kinetics (5 Marks)
- Factors affecting renal clearance of the drugs (5 Marks)
- Steady-state concentration (5 Marks)

4. (i) Sketch a fully labelled graphical representative of a dose response curve of an agonist drug (5mks)

(ii) Sketch a fully labelled graphical representative of a log dose response curve of an agonist drug. (5mks).

(iii) Sketch a log-Dose response relationship for three agonists A, B and C that vary in both potency and efficacy with the least potent having the highest efficacy (5mks).

(iv) Sketch a log dose response curve showing duality of a partial agonist. (5 marks)

(v) Define LD50 and ED 50? (1mark)

(vi) What is a therapeutic index? (1 marks)

(vii) State 3 benefits of using a sigmoid curve? (3marks)

Show a much larger range of concentrations

The portion of the curve between 20 and 80% of E_{max} is approximately a straight line.

It is easier to quantitative values

END